Identification and Control of the Two-Degree-of-Freedom Landed Helicopter (Heli2DoF)

Eric Castro Velásquez, Esteban Segura García, Daniel Chavarría García
School of Electronics Engineering
Instituto Tecnológico de Costa Rica (ITCR)
30101 Cartago, Costa Rica
{ecastro, essegura, dachavarria}@estudiantec.cr

Abstract—This paper presents the system identification and control design for a two-degree-of-freedom (2-DoF) landed helicopter platform (Heli2DoF). Four SISO transfer functions were identified from open-loop experiments and assembled into a MIMO state-space model in observable canonical form. Two integral state-feedback regulators (REI) were designed and validated experimentally: pole placement (PP) and Linear Quadratic Regulator (LQR). Both controllers satisfy the transient specifications ($t_{s_{2\%}} < 8\,\mathrm{s}$, $M_p < 5\%$). The LQR achieves $t_s = 2.92\,\mathrm{s}$ and $|\beta|_{\max} = 0.026\,\mathrm{rad}$, outperforming PP in coupling suppression and convergence speed; PP delivers a stronger disturbance rejection score (PERT 94/100 vs. 83/100).

Index Terms—system identification, state feedback, integral regulator, LQR, pole placement, MIMO

I. Introduction

The Heli2DoF platform is a landed two-degree-of-freedom helicopter whose pitch and yaw axes exhibit mechanical and aerodynamic cross-coupling. The objective of this work is to identify empirical SISO models for each input–output channel, assemble a MIMO state-space model, design integral state-feedback regulators via pole placement and LQR, and validate the designs experimentally. Section II describes the plant; Section III covers identification; Section IV presents the REI control design; Section V reports experimental results and a comparative analysis; Section VI discusses the findings, and Section VII concludes.

II. Plant Description

The platform consists of an aluminum pivot arm on a steel central axis. Two 24 V brushed DC motors driving GEMFAN 5045BN propellers actuate the pitch (θ) and yaw (β) axes through a dual H-bridge (24 V / 7 A). Angular position is measured with USDigital E5 optical encoders (3600 CPR, $k_\theta = 2\pi/3600 \approx 1.745 \times 10^{-3}$ rad/count). The PWM period is 500 µs at 24 V, and the sampling period is $T_s = 20\,\mathrm{ms}$. A Bluetooth link transmits the pitch encoder reading. The control inputs are the motor voltages $\mathbf{u} = [e_p,\, e_y]^T$ and the outputs are the angles $\mathbf{y} = [\theta,\, \beta]^T$.

III. System Identification

A. Experimental Protocol

Two open-loop experiments excited the plant: (i) a trapezoidal voltage up to 18.7 V applied to the pitch motor from $t = 1\,\mathrm{s}$, with a 6 V pulse on the yaw motor between $t = 12\,\mathrm{s}$ and $t = 20\,\mathrm{s}$ to counteract yaw drift; (ii) a 6 V rectangular pulse on the yaw motor with the pitch motor off. Only the first 10 s (PITCH) and 4 s (YAW and cross channels) were used for MATLAB ident fitting, and pre-trigger negative-time samples were removed.

B. SISO Transfer Functions and Parameters

The identified channels are

$$G_{11}(s) = \frac{k_p(s+z_p)}{s^2 + a_{1p}s + a_{0p}}, \quad G_{22}(s) = \frac{-k_y}{s(s+p_y)}, \qquad (1)$$

$$G_{21}(s) = \frac{k_{yp}}{s}, \quad G_{12}(s) \approx 0. \qquad (2)$$

The yaw-to-pitch coupling was found negligible. Identified parameters are in Table I.

TABLE I: Identified SISO Model Parameters

Parameter	Symbol	Value	Units
Pitch gain	k_p	0.01462	rad/(V·s^2)
Pitch zero	z_p	3.9330	rad/s
Pitch damping	a_{1p}	0.4450	rad/s
Pitch nat. freq.^{2}	a_{0p}	1.6410	rad^2/s^2
Yaw gain	k_y	0.03164	rad/(V·s^2)
Yaw pole	p_y	0.1000	rad/s
Cross gain	k_{yp}	0.1600	rad/(V·s)

C. MIMO State-Space Model

With $\mathbf{x}^T = [\theta\ \beta\ \dot\theta\ \dot\beta]$ and observable canonical form,

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_{0p} & 0 & -a_{1p} & 0 \\ 0 & 0 & 0 & -p_y \end{bmatrix}, \ \mathbf{B} = \begin{bmatrix} k_p & 0 \\ k_{yp} & 0 \\ k_p z_p & 0 \\ 0 & -k_y \end{bmatrix}, \qquad (3)$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \qquad (4)$$

The four original SISO transfer functions are recovered via zpk(modeloHeli) in MATLAB.

IV. Control Design

A. REI Structure

The augmented state is $\mathbf{x}_e^T = [\mathbf{x}^T \ \mathbf{x}_I^T]$ with $\dot{\mathbf{x}}_I = \mathbf{r} - \mathbf{y}$ and control law $\mathbf{u} = -\mathbf{K}\mathbf{x} - \mathbf{K}_i\mathbf{x}_I$. The extended system has six poles (four plant poles plus two integrator poles), guaranteeing zero steady-state error and active disturbance rejection. Design specifications: $t_{s_{2\%}} < 8\,$s, $M_p < 5\%$, $|\beta|_{\max} < 0.1\,$rad, $|e_{ss}| < 0.003\,$rad. Reference: $[\theta_r;\ \beta_r] = [0.583;\ 0]\,$rad (Simulink coordinates).

B. Pole Placement

A dominant complex conjugate pair was chosen to meet the transient specs; the remaining four poles were placed with real part at least $5\times$ that of the dominant pair:

$$\lambda^{PP} = \{-1.08 \pm j0.97,\ -4.32,\ -5.40,\ -6.49\}. \tag{5}$$

Computed gains via place(Ae,Be,p):

$$\mathbf{K}_{PP} = \begin{bmatrix} 32.94 & 16.14 & 15.04 & 7.18 \\ 14.14 & -81.14 & 38.19 & -75.43 \end{bmatrix}, \tag{6}$$

$$\mathbf{K}_{i,PP} = \begin{bmatrix} 15.28 & 8.55 \\ 15.99 & -26.71 \end{bmatrix}. \tag{7}$$

C. LQR

The cost function $J = \int_0^\infty (\mathbf{x}_e^T \mathbf{Q} \mathbf{x}_e + \mathbf{u}^T \mathbf{R} \mathbf{u})\,dt$ was minimized with $\mathbf{Q}$ and $\mathbf{R}$ tuned iteratively. The resulting gains are

$$\mathbf{K}_{LQR} = \begin{bmatrix} 27.76 & 2.89 & 16.21 & 1.15 \\ 2.76 & -53.25 & 2.98 & -41.78 \end{bmatrix}, \tag{8}$$

$$\mathbf{K}_{i,LQR} = \begin{bmatrix} 25.83 & 2.09 \\ 5.90 & -30.20 \end{bmatrix}. \tag{9}$$

V. Experimental Results

Both controllers were implemented on the Heli2DoF platform with $T_s = 20\,$ms. Aggregate metrics are summarized in Table II; Figs. 1–6 show the time responses for pitch tracking, yaw constraint, and disturbance rejection.

TABLE II: Experimental Comparison: PP vs. LQR

Ctrl.	Cond.	t_s (s)	M_p (%)	e_{ss}	$\lvert\beta\rvert_{\max}$	SC
PP	Sim.	4.50	2.42	0	—	—
	Exp.	5.12	3.12	0.333*	0.1010	93.9
LQR	Sim.	4.09	0.68	0	0.574	—
	Exp.	2.92	2.40	0.193*	0.0260	92.0
Spec.		< 8	< 5	< 0.003	< 0.1	—

*Linked to Simulink↔PASCAL coordinate transform.

A. Pole Placement Results

The Pole Placement controller achieved satisfactory tracking performance in the pitch axis, meeting all transient specifications with a settling time of $5.12\,$s and an overshoot of 3.12%. As observed in Figs. 1 and 2, the controller maintained acceptable yaw coupling, although the maximum yaw excursion reached $0.101\,$rad, which is essentially at the imposed constraint boundary. The disturbance rejection experiment shown in Fig. 3 demonstrates a rapid recovery after the perturbation event, resulting in the highest PERT score among the evaluated controllers (93.9). These results indicate that the pole placement design provides a robust and well-balanced performance, particularly in disturbance attenuation.

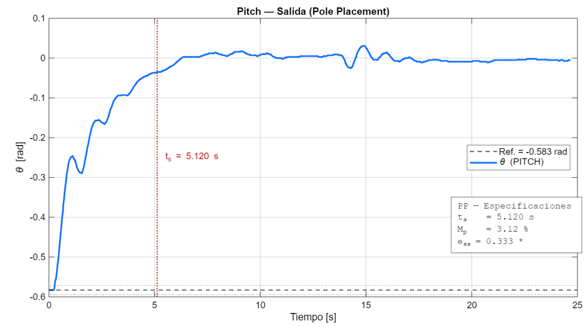

Fig. 1: Experimental pitch response using Pole Placement control. The transient specifications are satisfied with a settling time of $t_s = 5.12\,$s.

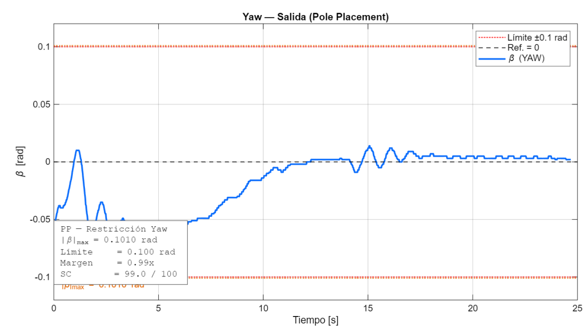

Fig. 2: Yaw axis response under the pitch step reference using Pole Placement control. The maximum yaw deviation reaches $|\beta|_{\max} = 0.101\,$rad, which is close to the specified limit.

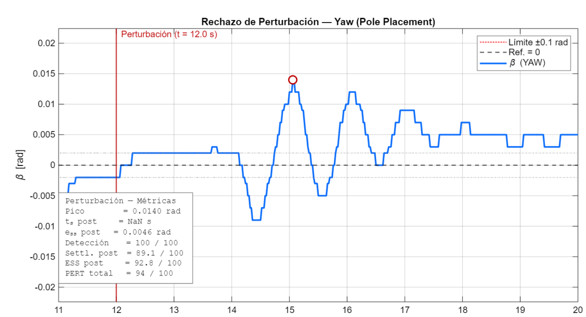

Fig. 3: Yaw disturbance rejection experiment using Pole Placement control. The disturbance event occurs at approximately $t = 12\,$s and achieves a total PERT score of 94/100.

B. LQR Results

The LQR controller exhibited the fastest pitch response, reducing the settling time to 2.92 s while maintaining a low overshoot of 2.40%, as shown in Fig. 4. Furthermore, the yaw response in Fig. 5 remained significantly below the specified limit, with a maximum deviation of only 0.026 rad, demonstrating superior decoupling between the pitch and yaw dynamics. Although the disturbance rejection performance in Fig. 6 yielded a slightly lower PERT score (92.0) than the Pole Placement controller, the recovery remained smooth and stable. Overall, the LQR design provided the best tracking speed and the largest yaw safety margin, highlighting the benefits of the optimal control formulation.

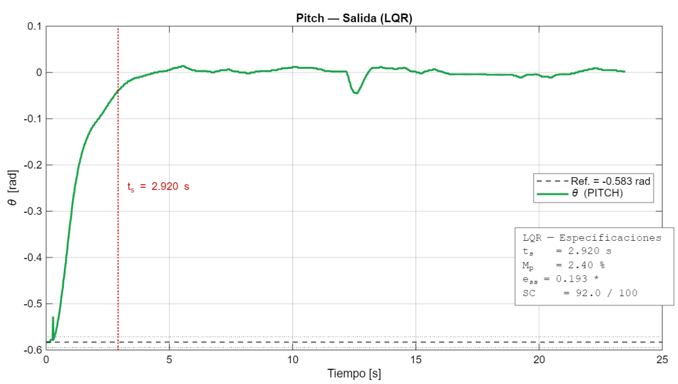

Fig. 4: Experimental pitch response using LQR control. The transient specifications are satisfied with a settling time of $t_s = 2.92\,\mathrm{s}$, providing faster convergence than Pole Placement.

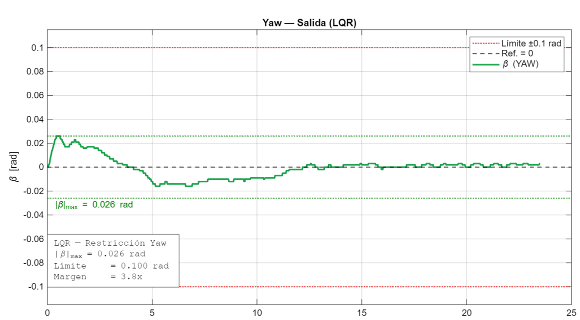

Fig. 5: Yaw axis response under the pitch step reference using LQR control. The maximum yaw deviation is limited to $|\beta|_{\max} = 0.026\,\mathrm{rad}$, providing approximately a 3.8× larger margin with respect to the specification.

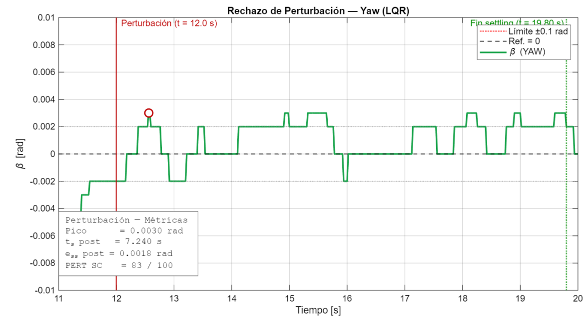

Fig. 6: Yaw disturbance rejection experiment using LQR control. The disturbance event occurs at approximately $t = 12\,\mathrm{s}$ and achieves a total PERT score of 83/100.

VI. Discussion

The LQR design improves settling time by approximately 30% relative to PP and reduces the peak yaw excursion by a factor of 3.8×, confirming superior coupling suppression. The simulated $|\beta|_{\max} = 0.574\,\mathrm{rad}$ exceeds the design specification because the linearized model overestimates the cross-coupling; the experimental peak (0.026 rad) is well within bounds.

The experimental steady-state error reported in IAE (0.193 for LQR, 0.333 for PP) is not a failure of the integral action—which yields $e_{ss} = 0$ in simulation—but reflects a constant offset introduced by the Simulink↔PASCAL coordinate transformation used during deployment. PP outperforms LQR in the disturbance rejection self-checker score (94/100 vs. 83/100), driven primarily by a higher ESS-post sub-score (92.8 vs. 53.7). Other unmodeled effects include actuator deadband and PWM quantization, nonlinear bearing friction, Bluetooth latency on the pitch encoder, and gyroscopic effects at high pitch velocity.

VII. Conclusion

Both REI designs satisfy the transient specifications experimentally. The LQR controller achieves $t_s = 2.92\,\mathrm{s}$, $M_p = 2.40\%$, and $|\beta|_{\max} = 0.026\,\mathrm{rad}$, providing the best overall transient and coupling-suppression performance. Pole placement reaches $t_s = 5.12\,\mathrm{s}$ with a marginal yaw excursion of 0.101 rad but a stronger post-disturbance score. Future work should correct the PASCAL coordinate offset to eliminate the residual experimental e_{ss}, retune $\mathbf{Q}/\mathbf{R}$ to improve the LQR ESS-post score, and explore LQG and H_∞ extensions for noise robustness and disturbance attenuation.

References

[1] E. Interiano, "Heli2DoF data and LabControl application," ITCR, 2020. http://www.ie.tec.ac.cr/einteriano/control/Laboratorio/4.5Heli2DoF.zip

[2] K. Ogata, Modern Control Engineering, 5th ed. Prentice Hall, 2010.

[3] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, Feedback Control of Dynamic Systems, 8th ed. Pearson, 2019.

[4] B. D. O. Anderson and J. B. Moore, Optimal Control: Linear Quadratic Methods. Prentice Hall, 1990.

[5] L. Ljung, System Identification: Theory for the User, 2nd ed. Prentice Hall, 1999.